`maxentcpp`: A C++ reimplementation of Maximum Entropy species distribution modeling for R.

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Abstract

`maxentcpp` is an R package offering a native C++17 reimplementation of the Maximum Entropy (Maxent) algorithm for species distribution modeling (SDM). The package translates the core continuous-predictor Maxent 3.4.4 fitting and projection algorithm into C++17 with R bindings via Rcpp, eliminating the Java Runtime Environment dependency entirely. It supports all five feature types, output transformations in raw, logistic, and cloglog scales, and a streaming raster evaluation engine for continental-scale projections. Numerical fidelity is validated per-iteration against the original Java implementation via a companion test package.

Summary

`maxentcpp` is an R package offering a native C++17 reimplementation of the Maximum Entropy (Maxent) algorithm for species distribution modeling (SDM). Maxent estimates the geographic distribution of a species from presence-only occurrence records and environmental covariates by identifying the probability distribution of maximum entropy subject to constraints derived from training data [21, 22]. Since its introduction, Maxent and its foundational methodological papers have been cited extensively across ecology, conservation biology, and biogeography [7].

The fitted model is a *Gibbs distribution* over geographic space:

$$P(\mathbf{x}) \;=\; \frac{1}{Z}\,\exp\Big(\sum_{j=1}^{J} \lambda_j\, f_j(\mathbf{x})\Big), \tag{1}$$

where f_j are feature transformations of the environmental variables, λ_j are learned weights, and $Z = \sum_{\mathbf{x}} \exp\big(\sum_j \lambda_j f_j(\mathbf{x})\big)$ is the normalizing constant (partition function). The optimizer finds the λ_j that maximize the regularized log-likelihood via sequential coordinate ascent with ℓ_1 (lasso) penalties [7, 21].

The original Maxent software is a Java desktop application [20]. `maxentcpp` translates the core continuous-predictor Maxent 3.4.4 fitting and projection algorithm into C++17 with R bindings via Rcpp [6] and RcppEigen [1], eliminating the Java Runtime Environment dependency entirely. The package supports all five feature types (linear, quadratic, product, hinge, and threshold), output transformations in raw, logistic, and complementary log-log (cloglog) scales, and a streaming raster evaluation engine that processes environmental layers block-by-block through the `terra`

package [12], enabling continental-scale projections on commodity hardware without loading entire raster stacks into memory. Tools for diagnosing model performance include AUC evaluation, permutation importance, response curves, and Multivariate Environmental Similarity Surfaces (MESS) for novelty detection.

Statement of need

Maxent is the *de facto* standard for presence-only species distribution modeling [17]. However, the original Java implementation presents a number of practical barriers for present-day ecological research workflows:

1. **Java dependency.** Java and `rJava` configuration can be a recurring source of installation and runtime failures, particularly across operating systems, Java versions, and managed computing environments. While `conda` environments and Docker containers can manage Java versioning, they add deployment complexity and are not standard practice in many ecology research groups.
2. **File-based I/O.** Data must be serialized to disk as SWD or ASCII grid files before the Java process can read them, and results must be parsed back from output files. There is no in-memory bridge between R data structures and the Java optimizer.
3. **Scalability.** The Java implementation is single-threaded and GUI-centric. In multi-species, multi-scenario workflows (e.g., 500 species $\times$ 4 climatic scenarios), file I/O overhead accumulates: each species run requires writing SWD files, invoking the JVM, and parsing output files. A native in-memory API eliminates this per-call overhead entirely.
4. **Maintenance.** The Java Maxent codebase has received only minor updates since version 3.4.4, with limited active development [20].

`maxentcpp` addresses these barriers by providing a native C++/R implementation. At present, the development version can be installed from GitHub using `remotes::install_github("alrobles/maxentcpp")`; following CRAN acceptance, the package will be installable with `install.packages("maxentcpp")`. The package operates entirely in memory through R objects and integrates seamlessly with the `terra` spatial data ecosystem. The intended users are ecologists, conservation biologists, and biogeographers who employ Maxent in reproducible, script-based workflows—from individual species analyses to large-scale biodiversity assessments.

Legacy software modernization in ecology

`maxentcpp` fits within a broader modernization trend in ecological and environmental-science software. Circuitscape.jl [10] illustrates how established ecological algorithms can be reimplemented in a modern high-performance language to improve scalability and parallel execution, while retaining continuity with a widely used landscape-connectivity tool. In R, the spatial ecosystem has similarly moved from older `sp`/`raster`-centered workflows toward `sf` [18] and `terra` [12], and SDM packages such as `ENMeval` [14] and `biomod2` [25] have followed this transition—with `ENMeval` additionally removing Java/`rJava` from its default path in favor of `maxnet`.

Within Maxent workflows specifically, existing modernization efforts address different parts of the problem. `rmaxent` [2] provides Java-free projection of previously fitted Maxent models from `.lambdas` files, parses feature weights and normalizers, and implements MESS-related diagnostics, but does not replace Java for model fitting. `maxnet` [19] provides a complete Java-free fitting implementation based on `glmnet` coordinate descent—preserving the same statistical model but employing a different optimization algorithm that can produce numerically different results, particularly for

threshold and hinge features.

`maxentcpp` occupies a distinct position within this landscape by porting the Java Maxent `density.Sequential` training optimizer itself to C++17 and validating optimizer trajectories against the Java implementation. To the best of current knowledge, `maxentcpp` is the first R package to target the Java Maxent 3.4.4 `density.Sequential` optimizer through a native C++ port and to publish per-iteration trajectory tests via the companion package `maxentcppCompTest` [8]. Its contribution is not modernization in general, but optimizer-level fidelity combined with integration into modern R/`terra` workflows.

State of the field

Several R packages provide access to Maxent-family models. **dismo** [13] is the most widely used R interface to Java Maxent; it wraps `maxent.jar` via `rJava`, inheriting the JDK dependency and file I/O overhead, and is currently in maintenance mode following the transition from `raster` to `terra`. **maxnet** [19] is a pure-R implementation that uses `glmnet` for elastic-net regularization; while it avoids the Java dependency, it employs a different optimization algorithm (elastic net via coordinate descent on the full feature matrix) rather than the sequential coordinate-ascent optimizer of the original Maxent, which can produce numerically different results, particularly for threshold and hinge features. **ENMeval** [14] is a model tuning and evaluation framework that wraps either `dismo::maxent()` or `maxnet::maxnet()`; it does not implement the Maxent algorithm itself. **kuenm** [3] is a calibration and evaluation toolkit that relies on `dismo` or direct Java Maxent calls.

Empirical comparison: maxentcpp vs maxnet

To quantify the practical differences between `maxentcpp` and `maxnet`, we compared both packages on a virtual species [16] with known true habitat suitability, constructed from the environmental layers bundled with `maxentcpp` (Annual Mean Temperature and Annual Precipitation over Central America, 2,371 non-NA cells). The virtual species has a Gaussian niche centered at 20 °C and 1,500 mm, and 100 presence records were sampled proportionally to the true suitability surface (see the companion vignette *Virtual Species Comparison*[1] for full reproducible code).

Both packages were fitted with linear, quadratic, and hinge features. In this illustrative example, the two packages produced highly similar rank and spatial-overlap patterns:

Table 1. Ecological equivalence metrics: `maxentcpp` vs `maxnet` on a virtual species.

Metric	Value
Schoener's D	0.93
Spearman ρ (rank correlation)	0.95
Pearson r	0.97
Mean $\lvert\Delta\text{cloglog}\rvert$	0.053
Max $\lvert\Delta\text{cloglog}\rvert$	0.43

These results suggest that `maxentcpp` and `maxnet` can produce highly similar predictions in a simple continuous-predictor setting (Schoener's $D > 0.9$ is conventionally interpreted as high niche overlap), while also showing non-negligible local differences in the tails of the suitability distribution, where the two optimization

[1] `https://alrobles.github.io/maxentcpp/articles/virtual_species_comparison.html`

algorithms (`maxentcpp`: sequential coordinate ascent; `maxnet`: elastic-net coordinate descent) regularize differently. A broader simulation study across multiple species, sample sizes, and predictor correlation structures would be needed to establish general equivalence.

The key practical scenarios where a user must choose `maxentcpp` over `maxnet` are: (1) when exact reproduction of Java Maxent results is required (e.g., replicating published analyses), (2) when lambda file interoperability with Java Maxent is needed, and (3) when streaming raster projection onto grids larger than available RAM is required.

Performance benchmarks

Training time for the bundled *Abeillia abeillei* dataset (73 presences, 2,371 background, linear + quadratic + hinge features), measured on an Intel Xeon VM (R 4.1.2, 100 `microbenchmark` repetitions with a fresh featured space per fit):

Table 2. Training time comparison.

Package	Median time per fit
`maxentcpp` (Sequential, default)	∼7.8 ms
`maxnet`	∼270 ms

After switching the default `maxent_fit()` backend from the legacy `goodAlpha` optimizer to the Java-faithful `Sequential` optimizer and vectorizing the hot loops with Eigen/BLAS, `maxentcpp` is now approximately 34 times faster than `maxnet` on this fixture, while preserving the per-iteration trajectory parity validated against Java Maxent 3.4.4. `maxnet` remains an excellent alternative when a pure-R `glmnet` path is preferred, but it uses a different optimization algorithm that can differ for threshold and hinge features. `maxentcpp`'s streaming evaluation also avoids materializing the full prediction matrix, which is expected to reduce peak memory during projection; future benchmarks should quantify this advantage on rasters larger than available RAM.

`maxentcpp` was built as a new package rather than contributing to existing projects for three reasons. First, a faithful port of the original Java optimizer (sequential coordinate ascent using ℓ_1-penalized Newton steps) requires a C++ engine that cannot be expressed through `glmnet`'s API; contributing this to `maxnet` would change its core algorithm. Second, `dismo`'s architecture is tightly coupled to `rJava` and `raster`; the native C++ strategy eliminates this dependency chain entirely. Third, the header-based C++ library design of `maxentcpp` enables reuse in other packages or standalone applications beyond R—something not achievable through modifications to existing R-only packages.

Software design

Architecture

`maxentcpp` is organized in three layers (C++ core, Rcpp bridge, R interface), with the C++ core further split into algorithmic and infrastructure sublayers:

The C++ core uses Eigen [9] for dense linear algebra. Of the ∼6 900 C++ lines, approximately 4 500 implement the core algorithmic logic (optimizer, features, density) while the remainder handles I/O and diagnostics. The high-level `maxent_run()` function provides a one-call workflow mirroring the Java GUI experience, while lower-level functions (`maxent_generate_features()`, `maxent_featured_space()`, `maxent_fit()`, `maxent_project_cloglog()`) offer full control over each modeling step.

Table 3. Architecture of `maxentcpp`: three layers with the C++ core divided into algorithmic and infrastructure sublayers.

Layer	Key components	Lines
C++ core (algorithmic)	`Sequential`, `FeaturedSpace`, five feature types	∼4 500
C++ core (I/O, diagnostics)	`BackgroundProvider`, grid I/O, CSV, MESS, response curves	∼2 400
Rcpp bridge	`rcpp_*.cpp` external-pointer bindings [5]	∼3 300
R interface	`maxent_run()`, feature generators, projection, evaluation, diagnostics	∼2 500

Optimization algorithm

The `Sequential` optimizer is a direct port of `density.Sequential` in Java Maxent 3.4.4, as represented by the public `mrmaxent/Maxent` repository and AMNH release notes (where 3.4.4 is described as a minor bug-fix release). It minimizes the ℓ_1-regularized negative log-likelihood:

$$\mathcal{L}(\boldsymbol{\lambda}) = -\frac{1}{m}\sum_{i=1}^{m}\sum_{j=1}^{J}\lambda_j\, f_j(\mathbf{x}_i) + \log Z(\boldsymbol{\lambda}) + \sum_{j=1}^{J}\beta_j\,|\lambda_j|, \tag{2}$$

where m is the number of presence records, $Z(\boldsymbol{\lambda}) = \sum_{k=1}^{N}\exp\!\big(\sum_j \lambda_j f_j(\mathbf{x}_k)\big)$ is the partition function summed over N background points, and $\beta_j = \hat{\sigma}_j/\sqrt{m}$ is the per-feature regularization parameter with $\hat{\sigma}_j$ being the standard deviation of feature j over the presence points (floored at 0.001).

At each iteration, the optimizer selects the feature j^* with the most negative $\Delta\mathcal{L}$ bound (the `deltaLossBound` function), then computes a Newton step:

$$\alpha_j = -\frac{\partial\mathcal{L}/\partial\lambda_j}{\mathbf{u}_j^{\top}\mathbf{H}\,\mathbf{u}_j}, \tag{3}$$

where $\mathbf{u}_j$ is the j-th coordinate direction and $\mathbf{u}_j^{\top}\mathbf{H}\,\mathbf{u}_j = \mathrm{Var}_q[f_j]$ is the variance of feature j under the current distribution q. The derivative includes the ℓ_1 subgradient: $\partial\mathcal{L}/\partial\lambda_j = \mathbb{E}_q[f_j] - \mathbb{E}_{\hat{p}}[f_j] + \beta_j\,\mathrm{sign}(\lambda_j)$. The step is damped during early iterations ($\alpha \leftarrow \alpha/50$ for iterations < 10, $\alpha/10$ for < 20, $\alpha/3$ for < 50) and falls back to a line search (`searchAlpha`) if the Newton step does not decrease loss. Every 10 iterations, a parallel update applies coordinate steps to all features simultaneously.

Convergence is tested every 20 iterations: training stops when the relative loss improvement falls below 10^{-5} or 500 iterations are reached. All constants in this section are taken directly from Java Maxent 3.4.4 source files, and each statement is covered by a source-mapping test in `maxentcppCompTest`.

Streaming raster evaluation

`maxentcpp` reads `terra::SpatRaster` objects block-by-block through a `BackgroundProvider` abstraction. Each tile is scored by the C++ engine and discarded before the next tile is loaded, allowing projection onto raster stacks larger than available RAM. The partition function Z is accumulated across tiles by summing unnormalized densities $\exp(\sum_j \lambda_j f_j(\mathbf{x}_k))$ for each cell k in the tile, then normalizing once after all tiles are processed. This produces results equivalent to single-pass evaluation because the mathematical sum is commutative and each cell is scored independently; floating-point summation order may introduce differences at the least-significant bit level.

Numerical fidelity

Each C++ class is a direct translation of its Java counterpart, preserving the same control flow, regularization logic, and numerical constants. This design choice prioritizes backward compatibility: users migrating from Java Maxent should obtain numerically equivalent results for implemented features under matched inputs, defaults, and deterministic settings. The fidelity contract is enforced by a dedicated companion package, `maxentcppCompTest` [8], which runs the C++ optimizer and the original Java `density.Sequential` on shared test fixtures and compares per-iteration trajectories of loss, entropy, and λ vectors.

On controlled test fixtures (identical input data, same floating-point representation), agreement reaches $< 10^{-14}$ relative error for symmetric fixtures and $< 10^{-6}$ on $\|\Delta\lambda\|_\infty$ for asymmetric fixtures at every iteration checkpoint. **These bounds hold under the specific conditions of the test fixtures** (same compiler family, IEEE 754 double precision, deterministic iteration order). Floating-point ordering differences introduced by alternative BLAS backends or aggressive compiler optimizations (e.g., `-ffast-math`) could widen the gap, though the sequential nature of the coordinate-ascent algorithm limits sensitivity to parallelism-induced reordering. Cross-platform CI (Ubuntu, macOS, Windows with GCC, Clang, and MSVC) confirms that the test fixtures pass on all three compiler families.

Table 4. Class-level porting map from Java Maxent to `maxentcpp`.

`maxentcpp` (C++)	Java Maxent original
`LinearFeature`	`density.LinearFeature`
`QuadraticFeature`	`density.SquareFeature`
`ProductFeature`	`density.ProductFeature`
`HingeFeature`	`density.HingeFeature`
`ThresholdFeature`	`density.ThresholdFeature`
`FeaturedSpace`	`density.FeaturedSpace`
`Sequential::run()`	`density.Sequential.run()`

Lambda file compatibility. Trained models are serialized in the same `.lambdas` text format used by Java Maxent 3.4.4. Lambda files produced by either implementation can be loaded by the other, ensuring interoperability with existing workflows and archived models.

Feature completeness

The following table summarizes the current scope of `maxentcpp` relative to Java Maxent 3.4.4 and `maxnet`. Legend: ✓ = implemented and tested; ∘ = partial or experimental; — = not implemented.

Categorical variables, replicate runs, jackknife, missing-data handling, and background bias weighting are now implemented and tested. SWD-to-raster projection remains a workflow convenience not yet exposed through the public R API; users can still project models by loading environmental grids via `terra` and the package's grid conversion functions.

Development provenance

The translation followed a phased approach across four publicly accessible repositories:

Table 5. Feature completeness: `maxentcpp` vs Java Maxent vs `maxnet`.

Feature	`maxentcpp`	**Java Maxent 3.4.4**	`maxnet` **0.1.4**
Linear features	✓	✓	✓
Quadratic features	✓	✓	✓
Product features	✓	✓	✓
Hinge features	✓	✓	✓
Threshold features	✓	✓	✓
Raw/logistic/cloglog output	✓	✓	✓
AUC evaluation	✓	✓	via ENMeval
Permutation importance	✓	✓	via ENMeval
Response curves	✓	✓	○
MESS maps	✓	✓	—
Clamping / fade-by-clamping	✓	✓	—
Lambda file I/O	✓	✓	—
Streaming raster projection	✓	—	—
Bias grids	✓	✓	via `offset`
Categorical variables	✓	✓	✓
Replicate runs / cross-val.	✓	✓	via ENMeval
Jackknife variable selection	✓	✓	—
Missing data handling	✓	✓	✓
SWD-to-raster projection	—	✓	—

1. `alrobles/Maxent`—a fork of the original `mrmaxent/Maxent` [20] Java source code, where the architecture was analyzed and each Java class was mapped to a C++ target (193 commits).
2. `alrobles/maxentcpp-devel`—the development repository where the C++ port, Rcpp bindings, R interface, tests, and documentation were iteratively built and refined (34 commits).
3. `alrobles/maxentcppCompTest`—the cross-language fidelity test suite containing Java oracle runners, golden trajectory files, and automated comparison scripts (40 commits).
4. `alrobles/maxentcpp`—the release repository with the clean, CRAN-ready package (47 commits).

Ecosystem comparison

The R ecosystem contains several packages that provide access to MaxEnt models, each employing a distinct integration strategy:

These packages can be grouped by their relationship to the MaxEnt algorithm:

- **Black-box wrappers** (dismo, kuenm, ENMTools, biomod2 MAXENT mode): invoke the Java binary and read output files. The optimizer is inaccessible.
- **Approximate reimplementations** (maxnet, kuenm2, biomod2 MAXNET mode): use `glmnet` coordinate descent on the logistic regression formulation. The statistical model is equivalent but the optimization path differs, which can produce numerically different results for threshold and hinge features.
- **Faithful reimplementation** (maxentcpp): ports the original Sequential optimizer to C++ and validates per-iteration numerical equivalence.

The distinguishing contribution of `maxentcpp` is that it is the only package offering a compiled native reimplementation of the actual MaxEnt `density.Sequential` optimizer—preserving both the statistical model and the optimization algorithm while

Table 6. R packages providing MaxEnt access.

Package	MaxEnt integration	Dependency
dismo [13]	rJava bridge to `maxent.jar`	Java JDK, rJava
kuenm [3]	`system2("java -jar maxent.jar")`	Java JDK
kuenm2 (Cobos et al.)	`glmnet` via forked `maxnet` code	glmnet
wallace [15]	Delegates to ENMeval → maxnet or dismo	ENMeval
ENMTools [28]	`dismo::maxent()` directly	dismo, rJava
biomod2 [25]	Java (`system2`) or `maxnet`	Optional Java or r
rmaxent [2]	Java-free projection of fitted Maxent models, `.lambdas` parsing, MESS	None (projection c
MIAmaxent [27]	MaxEnt via subset selection (not lasso)	glmnet
SDMtune [26]	Trains/tunes SDMs via dismo or maxnet	dismo or maxnet
maxentcpp	Native C++17 via Rcpp	None (self-contain

eliminating the Java dependency.

Limitations and inherited assumptions

The Maxent 3.4.4 algorithm that `maxentcpp` faithfully reproduces has well-documented limitations that users should be aware of:

- **Feature explosion.** The number of features (particularly hinge and threshold) grows with the number of environmental variables, which can cause identifiability issues in high-dimensional settings [7, 17].
- **Sensitivity to background sampling.** Maxent's output is influenced by the choice and size of the background sample, which affects the estimated density and can bias predictions in undersampled regions [**?**].
- ℓ_1 **regularization limitations.** The sequential coordinate ascent with ℓ_1 penalties produces sparse models but does not guarantee selection of the "correct" features under high correlation among predictors [11].
- **No principled uncertainty quantification.** Unlike Bayesian approaches or bootstrap-based methods, the Maxent optimizer produces point estimates without confidence intervals.

A principled alternative would be to reformulate the Maxent objective using modern convex optimization tooling (e.g., proximal gradient methods, ADMM) or Bayesian inference. However, such reformulations would produce different results than the widely used Java implementation, breaking comparability with published analyses. The design of `maxentcpp` explicitly aims to preserve this comparability.

CRAN compliance and software quality

Achieving CRAN acceptance requires more than passing `R CMD check`: packages must meet strict software-quality standards that safeguard user environments and ensure reproducibility across platforms [4, 24]. During the pre-submission review process, `maxentcpp` underwent targeted improvements to satisfy these requirements, detailed below.

Example documentation: `\dontrun{}` to `\donttest{}`

CRAN policy distinguishes between examples that *cannot* execute (missing external software, API keys) and those that are merely slow or resource-intensive. The former require `\dontrun{}`; the latter must use `\donttest{}` so that CRAN's check

infrastructure can optionally execute them with `--run-donttest` [23]. All 36 example blocks in `maxentcpp` were originally wrapped in `\dontrun{}`, which suppresses execution entirely and misleads users with a `# Not run:` comment. These were migrated to `\donttest{}`, converting the documentation examples into live, testable code that CRAN infrastructure can verify on every platform.

Self-contained examples with synthetic data

Changing the wrapper from `\dontrun{}` to `\donttest{}` revealed a deeper issue: most examples referenced undefined objects (e.g., a pre-fitted model `m`, a grid `g1`, or a results data frame) that only existed in the context of a vignette narrative. Under `\dontrun{}` this was invisible because the code was never evaluated; under `\donttest{}` it caused immediate failures.

Every example was rewritten to be self-contained, generating synthetic data at the point of use:

```
# Example: maxent_permutation_importance
n   <- 50
env <- data.frame(bio1 = runif(n, 10, 30),
                  bio12 = runif(n, 500, 2500))
f   <- maxent_generate_features(env, features = "lqh")
fs  <- maxent_featured_space(f, sample(n, 10))
m   <- maxent_fit(fs)
maxent_permutation_importance(m, fs)
```

This pattern—using `runif()`, `data.frame()`, and the package's own constructors (`maxent_grid_from_matrix`, `maxent_featured_space`, `maxent_fit`)—ensures that each example runs in isolation without relying on objects from other functions or prior sessions.

Graphics state restoration via `on.exit()`

CRAN requires that functions never leave the user's session in an altered state: any modifications to `par()`, `options()`, or the working directory must be reverted, even if the function errors mid-execution [24]. The `maxent_plot_response_curves()` function modified `par()` inside a loop to configure PNG output files but restored settings manually at the end of each iteration—a pattern that leaks state on error.

The fix extracts device management into an internal helper, `.maxent_safe_png()`, that encapsulates the open-device → set-par → plot → close-device lifecycle with guaranteed cleanup:

```
.maxent_safe_png <- function(file, width, height, expr) {
  grDevices::png(file, width = width, height = height)
  oldpar <- graphics::par(mar = c(4, 4, 2, 1))
  on.exit({ graphics::par(oldpar); grDevices::dev.off() },
          add = TRUE)
  force(expr)
}
```

Because `on.exit()` is function-scoped, placing the cleanup inside a dedicated helper avoids the anti-pattern of accumulating or overwriting exit handlers within a loop.

Optional dependency management

`maxentcpp` lists `terra` as a suggested (optional) dependency for raster I/O operations. CRAN checks may run with `_R_CHECK_FORCE_SUGGESTS_=false`, causing examples that assume `terra` is installed to fail. All terra-dependent examples are now guarded:

```
if (requireNamespace("terra", quietly = TRUE)) {
  # example code using terra
}
```

This ensures graceful skipping when the optional dependency is unavailable, following CRAN's recommended pattern for structuring examples that depend on suggested packages [23].

Continuous integration

GitHub Actions workflows run `R CMD check --as-cran` on Ubuntu (R release and R-devel), macOS (R release), and Windows (R release), with the `--run-donttest` flag enabled to exercise all example code. The CI matrix also includes a dedicated CRAN-preflight job that sets `_R_CHECK_FORCE_SUGGESTS_=false` to simulate CRAN's minimal-dependency environment.

Research impact statement

`maxentcpp` currently provides a numerically faithful implementation of the core continuous-predictor Maxent fitting and projection workflow for the implemented feature classes and output transformations. Recent releases add categorical predictors, replicate runs and cross-validation, jackknife diagnostics, and missing-data handling, closing most of the remaining gaps with Java Maxent 3.4.4 for the supported feature classes. SWD-to-raster projection remains a convenience workflow not yet exposed through the public R API.

The package is distributed under the MIT license (compatible with Eigen's MPL2 and RcppEigen's GPL-2 via the "or later" clause), with full source code, a pkgdown documentation website (`https://alrobles.github.io/maxentcpp/`), and four vignettes covering a getting-started walkthrough, the mathematical foundations of Maxent features, a comparative motivation document, and a virtual species validation study. Continuous integration via GitHub Actions runs `R CMD check` on Ubuntu, macOS, and Windows across R release and R-devel. The test suite comprises over 20 test files (∼4 800 lines) covering feature generation, optimization convergence, spatial projection, Java numerical equivalency, model diagnostics, and end-to-end workflows.

By providing a dependency-free, memory-efficient, and numerically faithful Maxent implementation, `maxentcpp` may lower the barrier for large-scale biodiversity assessments, climate-change projections, and conservation prioritization studies that rely on presence-only species distribution models. The recently added support for categorical predictors, replicate diagnostics, and missing-data handling brings the package closer to parity with Java Maxent 3.4.4 for the implemented feature classes.

AI usage disclosure

Generative AI tools were used during the development of `maxentcpp` and the preparation of this manuscript, as detailed below.

Tools used.

- **GitHub Copilot** (GPT-4-based, 2025 version): code scaffolding and boilerplate generation during the initial porting phases in `alrobles/Maxent` and `alrobles/maxentcpp-devel`.
- **Devin** (Cognition AI, 2025–2026): incremental feature implementation, test writing, documentation generation, CRAN compliance fixes, CI configuration, and manuscript drafting.
- **Claude** (Anthropic, 2025–2026): complex refactoring, cross-cutting documentation, and code review.

Nature and scope of assistance. AI tools contributed to: C++ code generation and refactoring from Java source, Rcpp binding scaffolding, R wrapper functions, `testthat` test scaffolding and expansion, `roxygen2` documentation, vignette drafting, pkgdown site configuration, CI workflow setup, CRAN compliance review, and manuscript drafting. The core algorithmic C++ classes (`Sequential`, `FeaturedSpace`, feature types) were translated from Java with AI assistance but under direct human supervision: each class was mapped line-by-line from the corresponding Java source, reviewed for correctness against the Java reference, and validated through the `maxentcppCompTest` trajectory comparison framework. Approximately 60% of the C++ algorithmic lines were initially drafted by AI tools and subsequently reviewed and corrected by the human author; the remaining 40% were written directly by the author, particularly the performance-critical optimizer inner loops and numerical edge cases.

Maintainability. The human author (Á. L. Robles Fernández) can independently explain and modify every algorithmically significant class. As evidence: the `Sequential::run()` optimizer, `good_alpha()`, `delta_loss_bound()`, `newton_step_feature()`, and `do_parallel_update()` methods are documented with line-by-line references to the corresponding Java source (e.g., "mirrors Sequential.java:294..342"), and the author has independently debugged numerical discrepancies during the fidelity testing process.

Confirmation of review. All AI-generated code, tests, documentation, and manuscript text were reviewed, edited, and validated by the human author (Á. L. Robles Fernández), who made all core design decisions including the algorithm translation strategy, API design, numerical fidelity requirements, and the phased development architecture. The full development history is publicly available in the four repositories listed above.

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